SUPPORTING ADULT GUIDANCE

Angles in Polygons

2D and 3D Shapes. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner finds the interior-angle sum and the interior and exterior angles of a polygon, and discovers two facts: the interior angles are fixed by the number of sides, and the exterior angles of any polygon always total 360°. The emphasis is on **reasoning the rule from the triangle split**, not memorising a formula.

Driving Question: *What decides the angles inside a polygon — and what stays the same, however many sides it has?*

The mathematics, in plain terms

Two facts drive this lesson. First, the interior angles of a polygon depend only on the number of sides: any polygon splits into $(n - 2)$ triangles, and each triangle holds 180°, so the interior angles add to **$(n - 2) \times 180°$**. For a *regular* polygon, each interior angle is that total shared equally, $(n - 2) \times 180° \div n$. Second, the **exterior** angles of any polygon always add to **360°** — walk once around the shape and you turn through exactly one full turn. Each exterior angle of a regular polygon is $360° \div n$, and at every corner the interior and exterior angle sit on a straight line, so they add to 180°. A regular polygon tessellates only when its interior angle divides exactly into 360° — which is why just the triangle, square and hexagon do.

Two fewer triangles than sides

The commonest slip is to multiply the number of sides by 180 — treating each side as a triangle. The single most useful habit is to draw the diagonals from one corner and actually count the triangles: there are always *two fewer* than the number of sides. Once a learner sees that, the formula $(n - 2) \times 180°$ stops being something to memorise and becomes something they can rebuild every time. The second habit is to hold the two facts side by side — interior angles depend on the number of sides, while the exterior angles always total 360° — and to notice that at each corner they meet on a straight line.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Filling in the table to discover that a polygon splits into $(n - 2)$ triangles, so its interior angles add to $(n - 2) \times 180°$.	~15 min

Movement	The Exterior-Angle Walk: stepping around a polygon and turning at each corner, to see the turns total exactly 360° — one full circle.	~8 min
Reflection	Judging true/false statements about polygon angles and, crucially, giving the reason why.	~10 min
Creativity	Investigating which regular polygons tessellate, and reasoning out why the angle must divide into 360°.	~5 min
Rest	A pause to notice which fact felt more surprising — the fixed interior angles, or the ever-present 360°.	~4 min
Nutrition	Contextual (Mode A): the honeycomb — why bees build hexagonal cells (they tile with no wasted space, and least wax for most honey).	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Using n × 180 for the interior sum

The learner treats each side as a triangle. Repair: “Draw the diagonals from one corner and count the triangles — is it the same as the number of sides, or two fewer?”

Thinking more sides means a bigger exterior angle

Repair: “Each exterior angle is 360° ÷ n. If n gets bigger, does 360 ÷ n grow or shrink?”

Confusing interior and exterior angles

Repair: “At one corner, mark the angle inside the shape, then the angle you turn through. They sit on a straight line — what must they add to?”

Believing the angles depend on the size drawn

Repair: “Draw a small and a large regular hexagon. Do the angles change, or only the side lengths?”

Assuming every regular polygon tessellates

Repair: “Does the interior angle divide exactly into 360°? Try 108° for a regular pentagon.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the angle sum of a triangle, angles on a straight line and around a point, and naming polygons. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the reasoning — and the triangle split — with the learner:

- “How many triangles does this polygon split into? So what is the angle sum?”
- “For a regular one, how would you share that total between the corners?”
- “What is each exterior angle — and how does it relate to the interior angle at the same corner?”
- “If you walked all the way around, how much would you have turned in total?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Do they need to memorise the formula?”
Understanding *why* — the polygon splits into (n – 2) triangles — matters more than memorising (n – 2) × 180°. If they can rebuild it from the triangle split, that is the goal.

“They keep using n × 180.”
Very common. Go back to the picture: draw the diagonals from one corner and count the triangles together — there are always two fewer than the number of sides.

“Why does the exterior-angle fact matter?”
It is often the quickest route (each exterior angle is 360° ÷ n), and it is a lovely invariant — true for every polygon, however many sides. It also explains why only three regular polygons tessellate.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Emerging → Developing → Secure → Mastering across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Emerging	Beginning to engage; names some features and makes a start with support.
Developing	Carries out the method with prompts; can explain part of the reasoning.
Secure	Works through independently and explains why each step is true.

Mastering	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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